

Intro to Creating Real-time Shaders

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ABSTRACT

Real-time shaders are incredibly powerful in game design. This hands-on session will introduce beginners to the basics of creating shaders for games and other real-time applications using a node-based shader editor. It will include a brief overview of the subject, step-by-step tutorials for creating three simple shaders (Distortion, Flipbook Animation, and Environmental Blend), as well as explanations of the math principles used. The session will be ideal for those wanting to get a jump start in learning shader creation. Though this course will be taught in Unity, the processes are very similar and can easily be translated to work inside of Unreal or other game engines with a node-based interface.

KEYWORDS

Computer Graphics, Real-time Shaders, Game Engine

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1 DISTORTION SHADER

We will start off with the creation of a distortion shader using noise textures to offset our UV coordinates. While a distortion shader may not be the most exciting shader, it is perfect to help understand the manipulation of UV coordinates. Using a grid texture we are able to show how the texture samples are affected by nodes such as the ADD, TIME, and MULTIPLY. We will work through the basics of UV manipulation including texture scaling with multiply and texture scrolling with addition to illustrate the real-time effect on the shader.

2 FLIP BOOK ANIMATION

Flip book effects are most commonly used by visual effects artists, and they are applied to particle billboards. Visual Effects Artists use them to represent very complex effects as cheaply as possible. One of the challenges in making a video game is everything has to be rendered in 33 milliseconds if you want to run at 30 frames per second or 16 milliseconds if you want 60 frames per second. That means there is very little time to do any complicated calculations. So visual effects artists often create complex effects offline and render them out into individual frames, arrange the frames in a

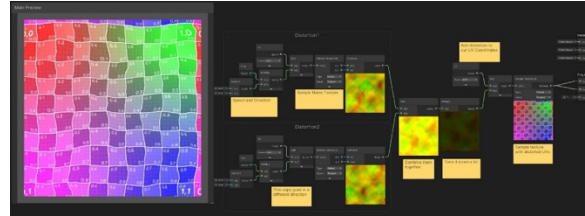


Figure 1: Sample Distortion node network

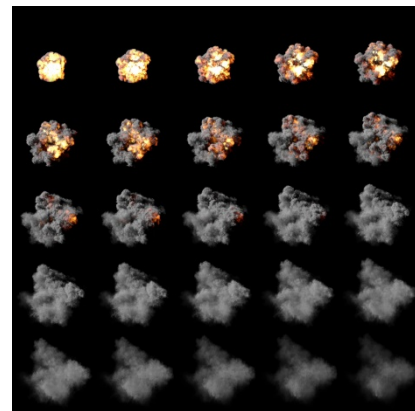


Figure 2: Sample grid texture of explosion

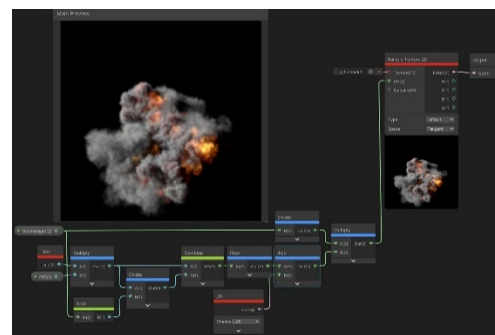


Figure 3: Example node network for flipbook effect shader.

texture atlas (as seen in figure 2), and then play back the frames and apply that texture to a particle. We start with a texture that is spit into a grid - where each tile in the grid represents one frame of animation.

Then we create a shader that plays back the grid tiles one at a time to create the animation. The tutorial covers both the use of the built-in flipbook node as well as a deeper explanation of how the math works for generating the effect.

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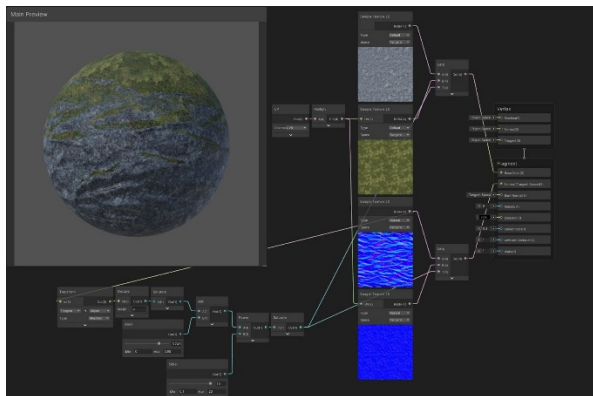


Figure 4: Example node network for mossy rock.

3 ENVIRONMENTAL BLENDING

The last type of shader we will look at is a method for creating a shader that allows you to apply environmental materials, like snow, sand, and moss to the top surfaces of your model. While this

technique is not new, it is used a lot to help save environment artists time and is something shader artists should learn how to do. In the past, artists were required to make one set of textures for mossy rocks, one set for snowy rocks, one for sandy and so forth for every different rock asset. But now we can create a single set of rock assets and change out the material that is applied depending on the environment they are in. We will achieve this effect by creating a mask using the world space normal of the model, and then use the mask to blend between the model's material and the environment material. You could use this shader to save time in asset creation - so you only need to create one set of assets that can be used in all environments.

4 CONCLUSION

The goal of this hands-on lab is to help beginners become familiar with the shader editor tool inside game engines (Unity/Unreal). They will become familiar with the techniques commonly used to create real-time shaders and gain a foundation to move to more complex and advanced effects. They will also be provided with notes on sources for continued learning.